Vaishanth Srinivasan

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• GitHub: https://github.com/VSLEGION

Summary

Aerospace engineer specializing in CFD, propulsion, and autonomous systems, with experience developing high-fidelity physics-based simulations and AI-driven optimization pipelines. Built and validated rotating detonation engine models and fuel-optimal UAV path planning algorithms, integrating first-principles engineering with scalable software systems. Focused on translating complex physical phenomena into deployable computational frameworks for real-world aerospace and autonomy applications.

Projects

ThermoSand – Interactive 2D Thermal Physics Simulation Sandbox with PINN Inference *(In Progress)*

- Developing real-time 2D heat equation solver with interactive Streamlit canvas, integrating classical finite-difference methods with a drawable material-geometry interface to simulate steady-state and transient temperature fields across user-defined boundary conditions and multi-material domains
- Implementing Physics-Informed Neural Network (PINN) surrogate enforcing PDE residual loss terms derived from the heat equation, replacing iterative numerical solves with learned inference while embedding physical constraints directly into the training objective
- Architecting modular solver pipeline progressing from FDM ground-truth baseline to live PINN-vs-classical comparison mode, with planned deployment on HuggingFace Spaces for GPU-backed real-time inference and solver benchmarking

Rotating Detonation Rocket Engine (RDRE) – Applied CFD and Propulsion Analysis

- Developed high-fidelity transient CFD model of a rotating detonation rocket engine in ANSYS Fluent using density-based solvers and UDF-coupled injectors to capture detonation–injection interactions under realistic conditions
- Modeled detonation across 8 fuel–oxidizer combinations using Chapman–Jouguet theory with a NASA CEA–MATLAB pipeline, predicting pressures up to ~300 atm and identifying ~15% improvement in pressure–temperature performance for optimal propulsion regimes
- Characterized unsteady detonation wave dynamics and pressure-gain combustion behavior, resolving injector coupling effects and complex flow physics in rotating detonation systems

CFD Analysis of Airfoil Icing Effects

- Developed high-fidelity CFD model of iced NACA 2412 airfoil in ANSYS Fluent using NASA/FAA-based rime and horn ice geometries (1–4% chord), with wall-resolved meshes ($y^+ \approx 1$) to capture separation and leading-edge flow behavior
- Quantified aerodynamic degradation across icing configurations, observing up to ~12% lift loss and >115% drag increase relative to clean baseline
- Identified transition to large-scale separation and bluff-body–like flow induced by horn ice, eliminating leading-edge suction peak; validated results against established icing literature

Dora the Pathfinder – Fuel-Optimal Path Finding

- Built modular Python-based UAV path planning system using A* and Dijkstra with fuel consumption as the optimization objective, replacing conventional distance-based routing
- Developed constraint-aware graph framework with dynamic flight corridors and no-fly-zone avoidance, modeling fuel burn using aerodynamic and propulsion relationships (lift, drag polar, power, SFC)
- Integrated wind-aware routing and path-smoothing to generate physically realistic trajectories, enabling energy-optimal navigation under varying environmental conditions

Experience

Artificial Intelligence and Augmented Reality Engineer ***April 2025 – July 2025***

Orange and Teal Pvt. Ltd., Bengaluru, Karnataka, India

- Architected and trained custom CNN models in PyTorch as part of an internal AI system for automating creative workflows, designing modular retraining pipelines to continuously improve model performance across evolving datasets and use cases
- Built end-to-end AI-driven marketing system integrating CRM data ingestion, customer behavior analytics, segmentation, and automated prompt generation to produce hyper-personalized advertisements with dynamic offer and discount logic tailored to individual user profiles
- Designed and deployed fully automated pipeline connecting analytics, model outputs, and delivery infrastructure (ad generation → email distribution), while optimizing system performance across accuracy, scalability, and inference latency through controlled experimentation

Conceptual Design Engineering Intern (New Product Development) ***April 2024 – August 2024***

TATA Advanced Systems Ltd., Bengaluru, Karnataka, India

- Extended and customized MDAO frameworks (OpenMDAO, OpenAeroStruct, OpenConcept) for conceptual aerospace vehicle design, replacing baseline isotropic assumptions with composite-aware structural models to improve fidelity of structural behavior and failure prediction
- Developed Python-based flight mechanics and stability analysis tools for constrained-deployment UAV systems, evaluating post-deployment dynamics and conducting longitudinal and lateral–directional stability studies across configuration variations
- Led aerodynamic trade studies on fluidic thrust vectoring and authored internal research on flapless flight concepts, presenting technical analyses and design trade-offs to senior engineering staff in formal reviews

Education

San José State University **San José, CA**

Master of Science in Aerospace Engineering ***August 2025 - Present***

GPA: 4.0

Relevant Coursework: AI systems for UAV pathfinding, CFD simulations of a custom Rotating Detonation Rocket Engine (RDRE), Analysis of Icing effects on aircraft airfoils using CFD

BMS College of Engineering **Bengaluru, Karnataka, India**

Bachelor of Engineering in Aerospace Engineering ***May 2024***

Skills

- **Simulation & Aerospace Engineering:** CFD (ANSYS Fluent) | Aerodynamics | Flight Mechanics | Stability Analysis | Propulsion Modeling | MDAO (OpenMDAO, OpenAeroStruct, OpenConcept)
- **Programming & Software Engineering:** Python (Advanced) | MATLAB | Algorithm Development | Modular System Design | Git
- **Machine Learning & AI:** PyTorch | Convolutional Neural Networks (CNNs) | Data-Driven Optimization | AI System Integration
- **Design & Visualization:** CATIA V5 | Unreal Engine 5 | Blender | Cinema 4D | Octane Render